

# Summary of Current Model Verification Study

## Goal

The goal of this round was to find a concrete fermionic model that can support the perturbative SDP improvement framework in the draft. In particular, we wanted to verify three points:

1. exactness of the reduced SDP at a reference Hamiltonian  $H_0$ ;
2. first-order response matching between the reduced SDP and the physical ground state at  $H_0$ ;
3. a nontrivial lower bound

$$\|\delta m_S(\theta)\| \geq c_{\text{mis}}|\theta|$$

for selected moments after a weak perturbation  $H(\theta) = H_0 + \theta V$ .

We tested two spinful Hubbard-type candidates, both based on an isolated Hubbard-dimer reference point.

## Common Setup And Notation

- Lattice: an open one-dimensional chain with  $L$  sites,  $L$  even.
- Filling: half filling,  $N_e = L$  electrons.
- Strong dimers:  $D_r = (2r - 1, 2r)$ ,  $r = 1, \dots, L/2$ .
- Weak inter-dimer bonds:  $B_r = (2r, 2r + 1)$ ,  $r = 1, \dots, L/2 - 1$ .
- Operators:
  - $c_{i,\sigma}, c_{i,\sigma}^\dagger$ : fermion annihilation/creation operators at site  $i$  with spin  $\sigma \in \{\uparrow, \downarrow\}$ .
  - $n_{i,\sigma} = c_{i,\sigma}^\dagger c_{i,\sigma}$ , and  $n_i = n_{i,\uparrow} + n_{i,\downarrow}$ .
  - $D_i = n_{i,\uparrow} n_{i,\downarrow}$ , the onsite double occupancy.
  - $K_{ij} = \sum_\sigma (c_{i,\sigma}^\dagger c_{j,\sigma} + c_{j,\sigma}^\dagger c_{i,\sigma})$ , the bond kinetic observable.
  - $S_i^z = (n_{i,\uparrow} - n_{i,\downarrow})/2$ ,  $S_i^+ = c_{i,\uparrow}^\dagger c_{i,\downarrow}$ , and  $S_i^- = c_{i,\downarrow}^\dagger c_{i,\uparrow}$ .
  - *h.c.* denotes the Hermitian conjugate.

The common reference Hamiltonian is the isolated-dimer spinful Hubbard model with  $t_d = 1$ :

$$H_0 = \sum_{r=1}^{L/2} [-K_{2r-1,2r} + U(D_{2r-1} + D_{2r})].$$

For response matching, fields are coupled to the selected moments  $M_1, \dots, M_s$  by

$$H(\theta, h) = H_0 + \theta V - \sum_{a=1}^s h_a M_a.$$

## Reported Quantities

The quantities in this subsection are diagnostics used in this numerical report, not standard names from the Hubbard-model literature.

- Reference energy error:  $|E_{\text{SDP}}(H_0) - E_{\text{phys}}(H_0)|$ .
- Selected-moment error:  $\|m_S^{\text{SDP}}(0) - m_S^{\text{phys}}(0)\|$ .
- Response-matching error:  $\|D_h m_S^{\text{SDP}}(0) - D_h m_S^{\text{phys}}(0)\|$ , the norm of the difference between the SDP and physical first-order response matrices.

- Mismatch ratio:  $\|\delta m_S(\theta)\|/|\theta|$ , where  $\delta m_S(\theta) = m_S^{\text{SDP}}(\theta) - m_S^{\text{phys}}(\theta)$ . This ratio is used to estimate  $c_{\text{mis}}$ .
- Solver: all reported SDP values below were computed with MOSEK.

## Terminology And References

- The terms “Hubbard model”, “spinful Hubbard chain”, “hopping”, “onsite interaction”, “half filling”, and “double occupancy” follow the standard Hubbard-model terminology; see Hubbard’s original paper and standard treatments of the one-dimensional Hubbard model [1, 2].
- The words “dimerized” and “alternating” refer to chains with alternating strong and weak bonds; this terminology is standard in the dimerized or alternating Hubbard-chain literature [3, 4].
- The term “extended Hubbard model” refers here to adding an intersite density-density interaction to the onsite Hubbard interaction; this is the standard usage in the extended-Hubbard literature [5].
- “Response matrix” is used in the usual static linear-response sense, following the Kubo linear-response terminology [6].
- “Semidefinite programming” follows the standard convex-optimization terminology of Vandenberghe and Boyd [7].

## Model 1: Standard Dimerized Spinful Hubbard Chain

**Model.** The first model is the standard alternating-hopping spinful Hubbard chain. It uses the common reference Hamiltonian  $H_0$  above and the weak inter-dimer hopping perturbation

$$V_{\text{hop}} = - \sum_{r=1}^{L/2-1} K_{2r,2r+1}.$$

The reported scan used  $L = 4$ ,  $U = 2$ ,  $t_d = 1$ , and  $\theta = \pm 0.005, \pm 0.01$ .

The selected moment families tested were:

- Bond kinetic moments on strong dimers:  $M_r = K_{2r-1,2r}$ .
- Double occupancy moments on strong dimers:  $M_r = D_{2r-1} + D_{2r}$ , and also the site-resolved versions  $D_{2r-1}$ ,  $D_{2r}$ .
- Spin correlation moments on strong dimers:  $S_{2r-1}^z S_{2r}^z$ , and  $\mathbf{S}_{2r-1} \cdot \mathbf{S}_{2r} = S_{2r-1}^z S_{2r}^z + \frac{1}{2}(S_{2r-1}^+ S_{2r}^- + S_{2r-1}^- S_{2r}^+)$ .
- Onsite-pair hopping moments between the two sites of a strong dimer:  $P_r = c_{2r-1,\uparrow}^\dagger c_{2r-1,\downarrow}^\dagger c_{2r,\downarrow} c_{2r,\uparrow} + h.c..$
- Singlet/triplet projectors on a strong dimer:  $\Pi_{s,r} = |s_r\rangle\langle s_r|$  and  $\Pi_{t0,r} = |t_r^0\rangle\langle t_r^0|$ , where  $|s_r\rangle = (|\uparrow, \downarrow\rangle - |\downarrow, \uparrow\rangle)/\sqrt{2}$  and  $|t_r^0\rangle = (|\uparrow, \downarrow\rangle + |\downarrow, \uparrow\rangle)/\sqrt{2}$  on dimer  $D_r$ .
- Bond kinetic moments on weak inter-dimer bonds:  $M_r = K_{2r,2r+1}$  on the weak bonds  $B_r$ .

## Verification results.

1. **Reference exactness.** The reference SDP was solved at  $H_0$ . The following rows give representative moment choices from the scan.

| Selected moments    | Energy error          | Moment error           | Status  |
|---------------------|-----------------------|------------------------|---------|
| $K_{2r-1,2r}$       | $1.47 \times 10^{-9}$ | $9.37 \times 10^{-8}$  | OPTIMAL |
| $D_{2r-1} + D_{2r}$ | $1.47 \times 10^{-9}$ | $4.74 \times 10^{-8}$  | OPTIMAL |
| $K_{2r,2r+1}$       | $1.47 \times 10^{-9}$ | $4.65 \times 10^{-16}$ | OPTIMAL |

2. **First-order response matching.** Strong-dimer moments had small response-matching errors. The weak-bond kinetic moment had a large response-matching error.

| Selected moments                          | Support             | Response error        |
|-------------------------------------------|---------------------|-----------------------|
| $K_{2r-1,2r}$                             | strong dimers $D_r$ | $4.86 \times 10^{-7}$ |
| $D_{2r-1} + D_{2r}$                       | strong dimers $D_r$ | $1.56 \times 10^{-7}$ |
| $\mathbf{S}_{2r-1} \cdot \mathbf{S}_{2r}$ | strong dimers $D_r$ | $1.44 \times 10^{-7}$ |
| $\Pi_{s,r}$                               | strong dimers $D_r$ | $1.10 \times 10^{-7}$ |
| $K_{2r,2r+1}$                             | weak bonds $B_r$    | 7.05                  |

3. **Mismatch lower bound.** For strong-dimer moments, the ratio  $\|\delta m_S(\theta)\|/|\theta|$  approximately doubled when  $|\theta|$  doubled from 0.005 to 0.01. This indicates quadratic, not linear, scaling in the tested range. The weak-bond kinetic moment had a stable linear ratio, but it failed the response-matching check above.

| Selected moments                          | Support       | Ratio at 0.005        | Ratio at 0.01         | Observation                         |
|-------------------------------------------|---------------|-----------------------|-----------------------|-------------------------------------|
| $K_{2r-1,2r}$                             | strong dimers | $5.96 \times 10^{-3}$ | $1.17 \times 10^{-2}$ | quadratic                           |
| $D_{2r-1} + D_{2r}$                       | strong dimers | $7.04 \times 10^{-3}$ | $1.40 \times 10^{-2}$ | quadratic                           |
| $\mathbf{S}_{2r-1} \cdot \mathbf{S}_{2r}$ | strong dimers | $6.36 \times 10^{-3}$ | $1.26 \times 10^{-2}$ | quadratic                           |
| $P_r$                                     | strong dimers | $5.42 \times 10^{-3}$ | $1.09 \times 10^{-2}$ | quadratic                           |
| $K_{2r,2r+1}$                             | weak bonds    | 4.59                  | 4.59                  | linear ratio,<br>response<br>failed |

## Model 2: Density-Perturbed Extended Spinful Hubbard Dimer Chain

**Model.** The second model uses the same isolated Hubbard-dimer reference  $H_0$ , but replaces weak hopping by an inter-dimer density-density perturbation:

$$V_{\text{dens}} = \sum_{r=1}^{L/2-1} n_{2r} n_{2r+1}.$$

The selected moments are the strong-dimer bond kinetic observables:

$$M_r = K_{2r-1,2r}, \quad r = 1, \dots, L/2.$$

The reported checks used  $U = 4$ ,  $t_d = 1$ , and  $L = 4, 6$ .

**Verification results.**

1. **Reference exactness.** At  $H_0$ , both the SDP energy and selected bond-kinetic moments matched the exact ground-state values to high precision.

| System         | Energy error           | Moment error          | Status  |
|----------------|------------------------|-----------------------|---------|
| $L = 4, U = 4$ | $5.74 \times 10^{-12}$ | $1.01 \times 10^{-7}$ | OPTIMAL |
| $L = 6, U = 4$ | $8.18 \times 10^{-13}$ | $1.03 \times 10^{-9}$ | OPTIMAL |

2. **First-order response matching.** The SDP and physical response matrices were compared by finite differences with field step  $h = 0.005$ .

| System         | Selected moments | Response error        |
|----------------|------------------|-----------------------|
| $L = 4, U = 4$ | $K_{2r-1, 2r}$   | $3.04 \times 10^{-5}$ |
| $L = 6, U = 4$ | $K_{2r-1, 2r}$   | $3.46 \times 10^{-6}$ |

3. **Mismatch lower bound.** The ratio  $\|\delta m_S(\theta)\|/|\theta|$  stayed nearly constant over the tested  $\theta$ -range, giving a numerical lower-bound constant.

| System         | Tested $ \theta $        | Ratio range | Conservative $c_{\text{mis}}$ |
|----------------|--------------------------|-------------|-------------------------------|
| $L = 4, U = 4$ | 0.002, 0.005, 0.01, 0.02 | 0.124–0.125 | 0.12                          |
| $L = 6, U = 4$ | 0.005, 0.01              | 0.215–0.216 | 0.21                          |

## Observed Status

For the standard hopping perturbation, the reference exactness error was small, the response check for moments supported on strong dimers passed numerically, and the corresponding mismatch was quadratic in  $\theta$  in the tested cases.

For the density-density perturbation, the tested  $L = 4$  and  $L = 6$  cases both gave small reference errors, small response-matching errors, and a linear mismatch ratio bounded away from zero on the tested  $\theta$ -range.

## References

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